

Ansh Pradhan

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PROFESSIONAL SUMMARY

Computer Engineering (AI Specialization) graduate with experience designing data-driven and AI-powered solutions across Machine Learning, Data Science, and Generative AI. Experienced in exploratory data analysis, predictive modeling, feature engineering, ML pipelines, LLMs, Retrieval-Augmented Generation (RAG), and AI Agents. Proficient in Python, SQL, TensorFlow, Scikit-learn, FastAPI, and LangChain, with a strong focus on developing scalable, production-ready intelligent applications.

EDUCATION

Institute of Advanced Research, Gandhinagar 2022 - 2026
B.Tech in Computer Engineering (AI Specialization) CGPA: 8.04/10

EXPERIENCE

- Institute for Plasma Research (IPR, Govt. of India)** | AI Research Intern *Feb 2026 – Jun 2026*
- Built Agentic RAG application using LangChain, FastAPI, LLMs with React(Vite) frontend with real-time token streaming.
 - Developed embedding pipelines, vector search workflows, and document ingestion systems using FAISS and Qdrant.
 - Implemented ReAct agents, retrieval pipelines, reranking, and knowledge graph integration for improved retrieval accuracy.
- Edunet Foundation** | AI/ML Intern *Jun 2025 - July 2025*
- Performed EDA & Developed XGBoost & Random Forest models on 48K+ records, improving prediction accuracy from 78% to 86%. Built an end-to-end ML pipeline including preprocessing, feature engineering, training, tuning, and evaluation.
 - Deployed a Streamlit-based prediction system for real-time inference and validation on 100+ test cases.
- India Space Academy** | Data Science Intern *Jun 2025 - July 2025*
- Performed ETL and EDA on 10+ astronomical datasets using Astropy for cosmological and galaxy cluster mass analysis.
 - Developed a real-time ISS tracker and 5+ scientific visualizations in Jupyter for research presentations.
 - Modeled 3D celestial coordinates using NumPy/Matplotlib with 95%+ spatial precision across large datasets.

- IOT-BASED STARTUP R&D Head** *Oct 2024 – Dec 2025*
- Leading the design and development of scalable IoT systems to address real-world challenges, ensuring seamless hardware-software integration.
 - Secured 1,00,000 SSIP funding for innovative, automation-driven product design with a focus on efficiency and impact.

TECHNICAL PROJECTS

- Chatur AI** (Individual) github.com/AnshPradhan14/Chatur_AI
- Built full-stack RAG assistant using FastAPI REST API, Groq LLM (LLaMA 3.1-8b), MongoDB Atlas vector database, and Tailwind CSS frontend.
 - Implemented JWT-authenticated multi-chat with PDF/URL ingestion, prompt engineering, & Mini-LM based embeddings.
 - Optimized inference pipeline by improving retrieval efficiency and reducing response latency using lightweight embeddings and hybrid search strategies; containerized via Docker for deployment on HuggingFace.
- Startup Idea Evaluator** (Individual) github.com/AnshPradhan14/startup-idea-evaluator
- Built a 3-agent CrewAI system (Market, Financial, Advisor) using Gemini 2.5 Flash and SerpAPI for startup validation.
 - Generated reports with market sizing, financial projections, risk analysis, and viability scores, reducing research time.
 - Developed a Streamlit interface for rapid idea evaluation, enabling consistent assessments across multiple startup concepts.
- ContractGuard Agent** (Individual) github.com/AnshPradhan14/ContractGuard-Agent
- Built AI-powered contract analysis assistant using FastAPI, LLMs, vector search, and intelligent document workflows.
 - Developed OCR pipelines, clause extraction systems, and contextual risk analysis workflows for sensitive legal documents.
 - Integrated AI agents, backend APIs, and retrieval pipelines for scalable compliance validation and secure document analysis.
- AeroMind: Aircraft Engine Predictive Maintenance Platform** (Individual) github.com/AnshPradhan14/AeroMind
- Built LightGBM/XGBoost models on NASA C-MAPSS data for RUL-estimation, anomaly detection, & engine health.
 - Implemented SHAP-based Explainable AI, quantile regression confidence intervals, & Streamlit dashboards for decisions.
 - Integrated a RAG-powered maintenance advisor using Llama 3 for actionable recommendations from model predictions.
- Resume Parser** (Individual) github.com/AnshPradhan14/Resume-Parser
- Built a FastAPI-based resume parsing system to extract structured data from PDFs using OCR and LLM-powered extraction.
 - Designed an end-to-end document processing pipeline integrating OCR, OpenAI APIs, and embedding generation for resume analysis.
- Advanced Age & Gender Detection** (Individual) github.com/AnshPradhan14/Age-and-Gender-Prediction
- Fine-tuned MobileNetV2 on 20,000+ UTKFace samples for demographic prediction with optimized preprocessing.
 - Deployed live inference via OpenCV for real-time performance; applicable for retail and security monitoring achieving 91% accuracy.

SKILLS

- **ML/AI:** Machine Learning, Deep Learning, EDA, ETL, Data Pipelines, NLP, LLMs, RAG, AI Agents, Prompt Engineering, Model Optimization, Vector Databases
- **Frameworks:** Numpy, Pandas, TensorFlow, PyTorch, Scikit-learn, LangChain, LangGraph, RestAPIs, FastAPI, Streamlit, Dockling, FAISS, ChromaDB, Qdrant, Computer Vision, YOLO, Ollama, Google ADK, CrewAI
- **Languages:** Python, SQL, C, C++
- **Tools:** Docker, Version Control - Git/GitHub, MongoDB Atlas, Power BI, N8N, OCR, API Integration(OpenAI, Anthropic, Gemini), Hugging Face
- **Familiar With:** MLOps, CI/CD, Kubernetes, GCP, System Design
- **Soft Skills:** Problem Solving, Communication, Fast Learning, Team Leadership, Mentoring

CERTIFICATIONS

- Supervised and Unsupervised Machine Learning - Stanford University (Coursera)
- Deep Learning - IBM (Coursera)
- IBM AI Fundamentals - IBM
- STTP - SpaceX Satellite Orbit Meta-Data Analysis with AI
- SAP Advance Course & SAP Value Added Course - Code Unnati

ACHIEVEMENTS & LEADERSHIP

- First Rank - Smart India Hackathon (SIH) Internal Round 2024
- Received Rs. 1,00,000 SSIP grant for an IoT-based smart system
- Vice-President, Computer Society & Gaming Club, IAR (July 2024 - June 2025)